

Physical Fitness and Health

体适能与健康



What is
“Physical Fitness?”
体适能

体适能 (Physical Fitness)

身体有足够的活力和精神应付日常工作外，还有能力享受余暇和应付突发事件。



“...the ability to carry out daily tasks with vigor and alertness, without undue fatigue and with ample energy to enjoy leisure-time pursuits and to meet unforeseen emergencies.”





Components of

Sports-Related

Physical Fitness



Components of Sports-Related Physical Fitness

Power (力量)

- 肌肉在最短时间收缩时所产生的最大张力
- the ability of a muscle or muscle group to exert great force in a very short period of time



Balance (平衡)

- 人体在静止站立或运动时能够维持身体稳定性的能力
- the ability of the body to maintain equilibrium while stationary or moving



Agility (灵敏)

身体或身体某部位迅速移动，并快速改变方向的能力

- the ability of the body to change directions quickly and accurately.



Components of Sports-Related Physical Fitness

Reaction time (反应时间)

- 身体对刺激做出迅速和准确反应的能力
- the ability of the body to react to a stimulus quickly and accurately.



Speed (速度)

- 人体进行快速移动的能力或最短时间完成某种运动的能力
- the ability of the body to move from one point to another point in a short period of time.

Coordination (协调)

- 肌肉系统表现的正确、和谐优雅的活动动作
- using the senses together (i.e. hand-eye coordination)



Components of Health-Related Physical Fitness

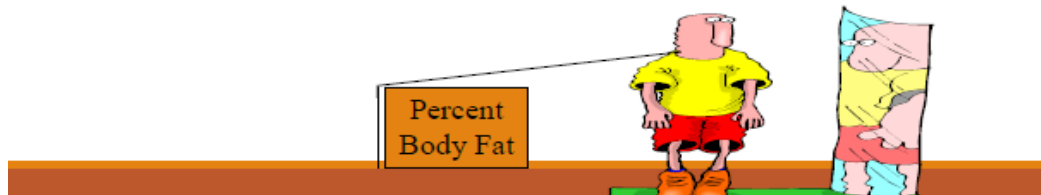
Cardiovascular endurance

- 包括心脏、肺脏、血管及血液等器官组织系统的机能，
- 指全身大肌肉进行长时间运动的持久能力，是体内心肺系统供养给身体各细胞及其用氧的能力。
- ability to continue or persist in strenuous tasks involving large muscle groups for extended periods of time



Body Composition

- 身体成分指组成人体组织器官的总成分
- The major chemical components of the body : fat mass (脂肪量), muscle (肌肉), bone (骨), and water volume (水)
- % body fat - preferred index used to evaluate a person's body composition



Muscular Strength & Endurance

Muscular Strength:

肌力是指肌肉一次收缩所发出的最大力量（肌力好，就能跑得快、跳得高、跳得远，提得动、掷得远）。



肌耐力是指某一肌群在从事反复收缩动作的持久力（跑步、跳绳、爬楼梯、后30秒钟的仰卧起坐）。

- maximal one-effort force that can be exerted against a resistance;

- Muscular Endurance :

- The ability of the muscles to perform repeated contractions without fatigue



Flexibility

- 指关节可以活动的最大范围。柔软性佳，关节活动范围大，协调性及灵活度就好，肌肉比较不易受伤，运动伤害也可以减少
- The ability of a joint to bend within the range of motion



- Prevent injury
- Prevent low-back pain
- Provides greater mobility

Neuromuscular Relaxation 神经肌肉松弛

- 身体能够有效放松或减轻不必要的紧张
- The ability to reduce or eliminate unnecessary tension or contraction in muscle groups.

运动金字塔 Motion pyramid

第一层:生活形态的体能运动 The first layer: the physical movement of life.

次数: 尽量每天数次 时间: 每项累计 30 分钟以上 强度: 适度。

Number of times: Try to be several times a day; Time: 30 minutes or more for each item;

Strength: Moderate.

这类活动主要包括走路、爬楼梯、骑车上班、园艺活动、家务、逛街、购物等。

Such activities mainly include walking, climbing stairs, cycling to work, gardening activities, housework, shopping, etc.

第二层:伸展运动 Second layer: stretching

次数: 每周 3-7 次; 时间: 6-10 个动作, 每个持续 30 秒; 强度: 伸展至有紧拉感

Number of times: 3-7 times a week. Time: 6-10 actions, each lasting 30 seconds; Strength: stretched to a tight feel.

这类运动主要包括瑜伽、拉筋动作、柔软体操等。

This type of exercise mainly includes yoga, lacing movements, and soft gymnastics.

第三层: 有氧运动和休闲运动 The third layer: aerobics and leisure sports

次数: 每周 3-5 次; 时间: 每次 20 分钟以上; 强度: 适度-激烈

Number of times: 3-5 times a week; Time: More than 20 minutes each time; Strength: Moderate to intense.

有氧运动有: 慢跑、骑自行车、游泳、登山、有氧舞蹈、健身操等; 休闲运动包括网球、篮球、高尔夫等球类运动。

Aerobics include: jogging, cycling, swimming, hiking, aerobic dance, aerobics, etc; leisure sports include tennis, basketball, golf and other ball games.

第四层:肌肉适能运动 The fourth layer: muscle fitness exercise

次数:每周 2-3 次; 时间:8-12 次为 1 组, 做 1-3 组; 强度:略超肌肉负荷。

Times: 2-3 times a week; time: 8-12 times for 1 group, do 1-3 groups; intensity: slightly more muscle load.

包括重量训练、仰卧起坐、俯卧撑、拉力带等。

Including weight training, sit-ups, push-ups, tension belts, etc.

第五层:静态活动 Fifth floor: static activity

尽量缩短减少, 不要连续超过 60 分钟

Try to shorten the reduction, not more than 60 minutes in a row.

这类活动包括看电视、玩电脑、工作等。

Such activities include watching TV, playing computer, working, etc.

**What is
“Health?”
健康**



传统观念: 没有疾病…… Traditional view: There is no disease

现代观念, 世界卫生组织 (WHO): 健康不只是没有疾病, 而且是一种在身体上、精神上 及社会和适应能力的完好状态。

Modern conception, world health organization (WHO) : health is not merely the absence of disease, but a state of physical, mental, and social fitness.

Contemporary view (WHO) – achievement of:

身体健康 Physical health

智能健康 Intellectual Wellness

情绪健康 Emotional Health

社交健康 Social Health

精神健康 Spiritual Health

Health and Avoidance of Disease

- 积极的健康与享受生活、挑战的能力有关

Positive health is associated with a capacity to enjoy life and withstand challenges.

- 消极的健康与发病率、过早死亡有关

Negative health is associated with morbidity and premature mortality.

US. Department of Health and Human Services (HHS). 2008. *Physical Activity Guidelines Advisory Committee report 2008*. www.health.gov/paguidelines/committeereprot.aspx

环境 Environment

- 自然因素 Physical factors

Air, Water, Pollution...

- 社会经济状况因素 Socioeconomic factors

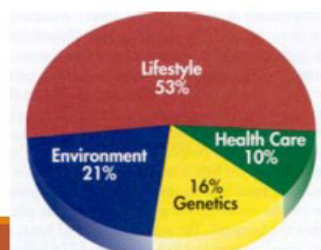
Income, housing(住房), employment (职业)...

- 家庭因素 Family

Parental values (父母价值观), divorce...

行为因素 Behavioral factors

- ◆ Smoking,
- ◆ Inactivity
- ◆ Fast driving
- ◆ Stress



Physical Activity and Health

What physical activity can do for health

- 降低慢性疾病 Reduces risks of many chronic diseases
- 控制体重 Promotes weight control
- 延迟早衰 Slows premature aging
- 提高生活质量 Improves quality of life



Physical activity is probably the best single thing a person can do for their health

（一）身体健康 Physical health

包括有充沛的体力，注意运动、营养和饮食的结合，不吸烟，不酗酒，不滥用药物，避免危险行为；留意自己的身体状况和由充足的休息。

Including ample physical strength, pay attention to the combination of exercise, nutrition and diet, no smoking, no alcohol, no drug abuse, avoid dangerous behavior; pay attention to your physical condition and adequate rest.

建议 Suggestion

一、运动习惯 exercise

1. 有氧运动（步行、跑步、游泳、跳绳……） Aerobic exercise (walking, running, swimming, skipping...)
2. 力量练习（哑铃、拉胶橡筋、徒手练习……） Strength exercises (dumbbells, rubber bands, hands-on exercises...)
3. 球类运动（篮球、羽毛球、网球……） Ball games (basketball, badminton, tennis...)
4. 每天 1 小时运动 1 hour of exercise per day

二、营养的摄取 Nutrition

1. 三餐正餐，定时定量 Three meals, regular timing
2. 晚餐在睡前 3 小时进食完毕 Dinner is finished 3 hours before bedtime
3. 尽量避免摄取高糖分、高脂肪的食物（糖果、朱古力、油炸食物、肥猪肉……） Try to avoid eating high-sugar, high-fat foods (candies, Chocolate, fried food, fat pork.....)
4. 尽量避免饮用酒精（每天不多于一罐啤酒） Try to avoid drinking alcohol (no more than one can of beer per day)
5. 不吸烟 No smoking

（二）智能健康 Intellectual Wellness

个人的智力和思维，在推理、处理问题和逻辑等方面的能力，也包括个人寻求知识和对新事物的接受和创作能力。

Personal intelligence and thinking, ability to reason, deal with problems and logic, as well as the ability of individuals to seek knowledge and accept and create new things.

建议 Suggestion

1. 每天都用一些时间去学习一些自己不急于需要的事物；
Take time every day to learn something you don't need in a hurry;
2. 经常留意社会问题及其变化；
Always pay attention to social issues and their changes;

3. 经常关注国际事态; Always pay attention to international events;

4. 对生活抱积极和肯定的态度, 遇有不如意或困惑, 不要让它缠绕自己多于 5 分钟;

Have a positive and affirmative attitude towards life. If you are unhappy or confused, don't let it wrap yourself for more than 5 minutes.

(三) 社交健康 Social Wellness

个人与家人、朋友、同学和同事等有良好的沟通关系, 乐于一同参与活动, 困难时相互扶持。

Individuals have good communication with family, friends, classmates and colleagues, and are willing to participate in activities together.

建议 Suggestion

1. 建立数位知己和密友; Create digital confidants and close friends;

2. 培养多种兴趣, 利于与人交往;

Cultivate a variety of interests and facilitate interaction with people;

3. 培养与人良好的沟通; Cultivate good communication with people;

4. 经常花时间去帮助及关心他人; Take time to help and care for others;

(四) 情绪健康 Emotional Wellness

能够适当地表达自己的感受、情绪与爱心, 能以冷静、理性和客观的态度去处理困难与挫折, 能充满信心地区面对挑战 and 逆境。情绪不稳定时, 也会懂得寻求帮助。

Being able to properly express one's feelings, emotions and love can handle difficulties and setbacks in a calm, rational and objective manner, and face challenges and adversity in areas of confidence. When you are emotionally unstable, you will also know how to seek help.

(五) 精神健康 Spiritual Wellness

个人能够找到生命的意义和目标, 包括个人价值观念、道德 观念、处事原则、信任、基本的求生能力和爱与被爱的能力等。精神健康的人能够欣赏大自然的奥妙, 并可与大自然融 为一体, 使个人内在精神充满动力, 迈向优质的人生。

Individuals can find the meaning and goals of life, including personal values, moral values, principles of doing things, trust, basic ability to survive, and the ability to love and be loved. People with mental health can appreciate the mystery of nature and integrate with nature, so that the inner spirit of the individual is full of motivation and a better life.

(六) 职业健康 Vocational Wellness

在工作上能够发挥个人才能，不断学习及提升自己的工作能力而获得工作的满足感，并能因自己的工作对社会做出的贡献感到光荣自豪。

I am able to use my personal talents at work, to continue to learn and improve my ability to work, to be satisfied with my work, and to be proud of my team's contribution to society.

推动健康人生的巨轮 **Promoting a healthy life**

健康人生是一多元化、富有动力的生活取向。目标清晰地利用自身的责任感可令生活更具有质量和意义。

A healthy life is a diverse and dynamic lifestyle. A clear use of your sense of responsibility can make life more quality and meaningful.

包含五种生活习惯(身体适能、个人习惯、心脏健康、营养与体重控制及压力处理)对我们的身心健康有莫大的影响。

Containing five lifestyle habits (physical fitness, personal habits, heart health, nutrition and weight control, and stress management) have a major impact on our physical and mental health.